

Education

Seoul National University	Mar 2020 – Feb 2027 (Expected)
Ph.D. in Computer Science and Engineering – Human-Centered Computer Systems Lab	
Advised by Youngki Lee	Seoul, Republic of Korea
Seoul National University	Mar 2018 – Feb 2020
M.S. in Computer Science and Engineering – Computer Graphics & Image Processing Lab	
Advised by Yeong-Gil Shin	Seoul, Republic of Korea
DigiPen Institute of Technology	Mar 2012 – Apr 2016
B.S. in Computer Science in Real-Time Interactive Simulation	Redmond, Washington, USA
DigiPen Institute of Technology Europe	Sep 2013 – Dec 2013
Exchange Semester	Bilbao, Spain

Work Experience

Microsoft Research Asia	Dec 2020 – May 2021
Research Intern — Heterogeneous and Extreme Computing (HEX) Group	
Advised by Yunxin Liu and Ting Cao	Beijing, China (Virtual)
Microsoft	Sep 2016 – May 2017
Software Engineer — Minecraft: Education Edition Team	Redmond, Washington, USA

Publication ([Google Scholar](#), †: Equal Contribution)

SpatialBlend: Relational Reconstruction for Cross-Space Presence in Heterogeneous Spaces	Nov 2026 (to appear)
Jingyu Lee, Dongho Han, Jinmyeong Kim, Youngki Lee	
ACM Symposium on User Interface Software and Technology (UIST)	
How Physics Fidelity Shapes Perceived Plausibility in Mixed Reality: A Case Study of Bounce Behavior	Oct 2026 (to appear)
Hyunsoo Kim, Jingyu Lee , Rajesh Balan, Youngki Lee	
IEEE Intl. Symposium on Mixed and Augmented Reality (ISMAR)	
Probing Ambient Co-Presence as Social Infrastructure: First Encounters with Connected Rooms	June 2026
Jingyu Lee, Youngki Lee	
ACM Intl. Workshop on Social Infrastructure Systems (SocialSys, with BuildSys) [DOI]	
Logan: Loss-tolerant Live Video Analytics System	Nov 2024
Kichang Yang, Minkyung Jeong, Juheon Yi, Jingyu Lee , Kyungsoo Park, Youngki Lee	
ACM Intl. Conference on Mobile Computing and Networking (MobiCom) [DOI]	

Maestro: The Analysis-Simulation Integrated Framework for Mixed Reality June 2024
Jingyu Lee, Hyunsoo Kim, Minjae Kim, Byung-Gon Chun, Youngki Lee
 ACM Intl. Conference on Mobile Systems, Applications, and Services (MobiSys) [DOI]

Band: Coordinated Multi-DNN Inference on Heterogeneous Mobile Processors June 2022
 Joo Seong Jeong[†], **Jingyu Lee**[†], Donghyun Kim, Changmin Jeon, Changjin Jeong, Youngki Lee, Byung-Gon Chun
 ACM Intl. Conference on Mobile Systems, Applications, and Services (MobiSys) [DOI]

ParallelFusion: Towards Maximum Utilization of Mobile GPU for DNN Inference June 2021
Jingyu Lee, Yunxin Liu, Youngki Lee
 ACM Intl. Workshop on Embedded and Mobile Deep Learning (EMDL, with MobiSys) [DOI]

Liver Segmentation in Abdominal CT Images via Auto-Context Neural Network and Self-supervised Contour Attention Mar 2021
 Minyoung Chung, **Jingyu Lee**, Jeongjin Lee, Yeong-Gil Shin
 Artificial Intelligence in Medicine [DOI]

Automatic Registration between Cone-Beam CT and Scanned Surface via Deep-Pose Regression Neural Networks and Clustered Similarities Dec 2020
 Minyoung Chung, **Jingyu Lee**, Wisoo Song, Youngchan Song, Il-Hyung Yang, Jeongjin Lee, Yeong-Gil Shin
 IEEE Transactions on Medical Imaging [DOI]

Deeply Self-Supervised Contour Embedded Neural Network Applied to Liver Segmentation Aug 2020
 Minyoung Chung, **Jingyu Lee**, Minkyung Lee, Jeongjin Lee, Yeong-Gil Shin
 Computer Methods and Programs in Biomedicine [DOI]

Pose-Aware Instance Segmentation Framework from Cone Beam CT Images for Tooth Segmentation May 2020
 Minyoung Chung, Minkyung Lee, Jioh Hong, Sanguk Park, Jusang Lee, **Jingyu Lee**, Il-Hyung Yang, Jeongjin Lee, Yeong-Gil Shin
 Computers in Biology and Medicine [DOI]

Publication (Preprint / Under Review)

Explorable Ideas: Externalizing Ideas as Explorable Environments Dec 2025
 Euijun Jung, **Jingyu Lee**, Minji Kim, Youngki Lee
 arXiv:2512.17017 [DOI]

Project

Band: Multi-DNN Framework for Mobile Platform
 Project Lead & Main Developer [website] [github] [UE plugin] [examples]

Somnus (Senior year student project at DigiPen)
Technical Director (Team of 4 Programmers, 1 Designer, 4 Artists, 3 Sound Designers)
3D Puzzle Platformer using perspective illusions [\[link\]](#)

Honors & Award

E3 College Game Competition — Somnus	Jun 2016
Showcased at Los Angeles Convention Center [DigiPen news] [Interview]	Finalist (top 5 out of ~400)
Bisa Scholarship	Mar 2012
Full-ride scholarship granted from DigiPen - KMU dual degree program	~\$60,000

Teaching

SNU & LG Data Science Course: Machine Learning	Jan 2021 - 2024, Feb 2020
Machine Learning Theory and Practice	Teaching Assistant
SNU & Samsung Data Science Course: Machine Learning	Oct 2023, Jun 2023
Machine Learning Projects with Kaggle	Teaching Assistant
SNU & SK Hynix Machine Learning Engineer Course	Nov 2022, Aug 2022
Time Series Data Analysis	Teaching Assistant
SNU Big Data & Fintech Expert Training Course	Apr 2021
Mobile Deep Learning & Machine Learning Application Development	Teaching Assistant
SNU Computer Programming (M1522.000600, 001)	Fall 2020
Object Oriented Programming in Java & C++	Teaching Assistant

Mentoring

Heeseo Kim	Jul 2025 – Feb 2026
Undergrad Research Intern, Human-Centered Computer Systems Lab	
XR Agents	
Junhee Kim	Jul 2025 – Present
Undergrad Research Intern, Human-Centered Computer Systems Lab	
XR Agents	
Euijoon Jung	Feb 2025 – Sep 2025
Ph.D. Student, Human-Centered Computer Systems Lab	
Idea Island (XR Ideation Tool)	
Jinmyeong Kim	Jun 2024 – Present
Ph.D. Student, Human-Centered Computer Systems Lab	
SpatialBlend ; XR Perception Platform (On-device Hand-Object Perception)	

Dongho Han

Jun 2024 – Present

Ph.D. Student, Human-Centered Computer Systems Lab

[SpatialBlend](#); XR Perception Platform (On-device Hand-Object Perception)

Hyunsoo Kim

Mar 2022 – Present

Undergrad Research Intern → Ph.D. Student, Human-Centered Computer Systems Lab

[Maestro](#); [MR Physics Fidelity](#)

Invited Talk

University of Texas at Austin

Dec 2024

Analysis-Simulation Integrated Platform For Mixed Reality

Host: Daehyeok Kim

Duke University

Dec 2024

Analysis-Simulation Integrated Platform For Mixed Reality

Host: Maria Gorlatova

University of California, Merced

Dec 2024

Analysis-Simulation Integrated Platform For Mixed Reality

Host: Shijia Pan

KAIST School of Computing

Sep 2023

Band: Coordinated Multi-DNN Inference on Heterogeneous Mobile Processors

Host: Junchwa Song

Samsung Global Technology Symposium — Poster

Apr 2023

Band: Coordinated Multi-DNN Inference on Heterogeneous Mobile Processors

Institute of Embedded Engineering of Korea Fall Conference Proceedings

Nov 2022

Band: Coordinated Multi-DNN Inference on Heterogeneous Mobile Processors

A3 Foresight Workshop

Aug 2021

ParallelFusion: Towards Maximum Utilization of Mobile GPU for DNN Inference [\[link\]](#)

Academic Service

IEEE Transactions on Vehicular Technology

Reviewer

IEEE Transactions on Medical Imaging

Reviewer

updated: June 28, 2026